



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: VIMAL R

Reg. No.: 192421350

Course Code: CSA5115

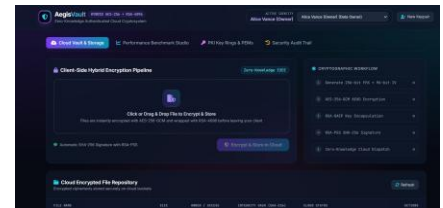
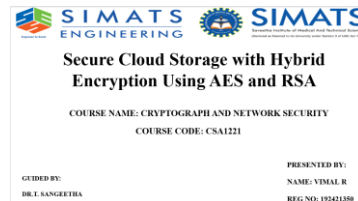
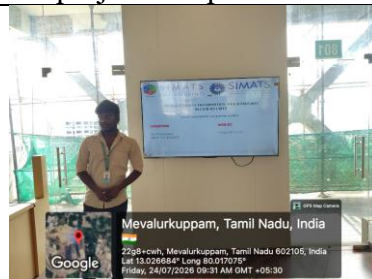
Slot: A

Course Name: CRYPTOGRAPHY AND NETWORK SECURITY

Course Faculty: DR. T. SANGEETHA

Technical Title: Secure Cloud Storage with Hybrid Encryption Using AES and RSA

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Technical Description: Development of a secure cloud storage system using AES-256 for file encryption and RSA-2048 for secure encryption key protection.

Secure Cloud Storage with Hybrid Encryption Using AES and RSA is a project designed to protect confidential files stored in a cloud-based environment using hybrid cryptography techniques. The system combines symmetric and asymmetric encryption to provide efficient and secure data protection. User authentication is implemented through registration, login, and password verification to ensure that only authorized users can access the system. When a user selects a file, an AES-256 key is generated and used to encrypt the file, providing fast and secure protection for large data. The generated AES key is then encrypted using the RSA-2048 public key, ensuring secure key distribution and management. The encrypted file and protected AES key are securely stored in the cloud storage system. During download, the authorized user's RSA private key is used to recover the AES key, which is then used to decrypt the file and restore the original data. The system improves cloud data confidentiality, secure key management, and controlled file access while providing a practical implementation of hybrid encryption for secure cloud storage.

Student Signature

Guide Signature